1. $3y - 7 - 5y + 4 + 4y - 2$
$= 3y - 5y + 4y - 7 + 4 - 2$
$= \mathbf{2y - 5}$

2. $\mathbf{-1.5 < x \leqslant 8}$

EXAM TIP
The numbers are greater than -1.5, but less than or equal to 8.

3. **(a)** Total amount $= \frac{18}{100} \times \$2250 + 24 \times \$92.75$
$= \mathbf{\$2631}$

(b) Required percentage $= \frac{2631 - 2250}{2250} \times 100\%$
$= \mathbf{16.9\%}$ (to 3 s.f.)

4. **(a)** Median time $= \mathbf{15.5\ h}$

(b) About 25% of the adults spent more than 20 hours (upper quartile).
About 25% of the adults spent less than 12.5 hours (lower quartile).
$\therefore$ Rishi is **incorrect**.

EXAM TIP
Identify the upper quartile and the lower quartile from the box-and-whisker plot first.
About 25% of the adults spent less than 12.5 hours.
About 75% of the adults spent less than 20 hours.

5. $$\frac{7x}{6} - \frac{3(x+1)}{8} - \frac{7x-6}{24} = \frac{4(7x) - 9(x+1) - (7x-6)}{24}$$
$$= \frac{28x - 9x - 9 - 7x + 6}{24}$$
$$= \frac{12x - 3}{24}$$
$$= \frac{3(4x-1)}{24}$$
$$= \mathbf{\frac{4x-1}{8}}$$

EXAM TIP
Combine the fractions into a single fraction with denominator 24 (LCM of 6, 8 and 24).
Then write $\frac{12x-3}{24}$ in its simplest form.

6. **(a)** 1 cm represents 200 000 cm $= \frac{200\ 000}{100 \times 1000}$ km
$= 2$ km.
$\therefore n = \mathbf{2}$

(b) Actual distance $= 18.9 \times 2$
$= \mathbf{37.8\ km}$

(c) 2 km is represented by 1 cm.
4 km^2 is represented by 1 cm^2.
Area on the map $= \frac{728.6}{4}$
$= \mathbf{182.15\ cm^2}$

EXAM TIP
If the linear scale of a map is $1 : r$, its area scale is $1 : r^2$.

7. **(a)** $18a - 24b + 15c = \mathbf{3(6a - 8b + 5c)}$

(b) $3 + 2m^2xy - 2my - 3mx$
$= 3 - 3mx + 2m^2xy - 2my$
$= 3(1 - mx) + 2my(mx - 1)$
$= 3(1 - mx) - 2my(1 - mx)$
$= \mathbf{(1 - mx)(3 - 2my)}$

EXAM TIP
Factorise by grouping.

8. Since the difference between any two consecutive terms is the same number, x is the average of the five terms.
$5x = 105$
$x = 21$
Common difference $= 21 - 15$
$= 6$
$\therefore$ $\mathbf{w = 9, x = 21, y = 27, z = 33}$

EXAM TIP
The sequence is symmetrical about x.
$x = \frac{15 + y}{2} = \frac{w + z}{2}$

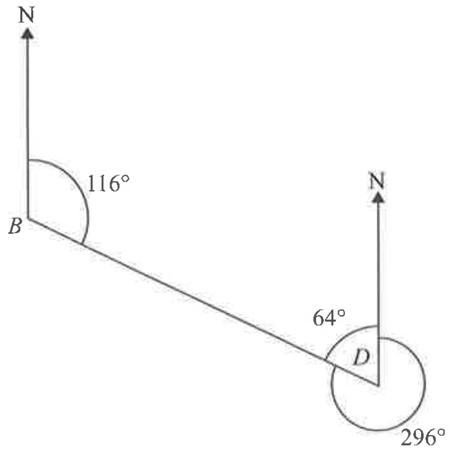
9. Shaded area $= \frac{96°}{360°} \times \pi(90)^2 - \frac{96°}{360°} \times \pi(24)^2$
$= \mathbf{6300\ cm^2}$ (to 3 s.f.)

EXAM TIP
Area of sector $= \frac{x°}{360°} \times \pi r^2$

10. **(a)** $\sin 58° = \frac{AC}{36}$
$AC = 36 \sin 58°$
$= \mathbf{30.5\ m}$ (to 3 s.f.)

(b) $\cos 58° = \frac{AB}{36}$
$AB = 36 \cos 58°$
$= 19.007$ m (to 5 s.f.)
Perimeter of $\triangle ABC = 36 + 19.077 + 30.530$
$= \mathbf{85.6\ m}$ (to 3 s.f.)

(c)



$180° - 116° = 64°$

$360° - 64° = 296°$

$\therefore$ Bearing of B from D is **296°**

EXAM TIP

Bearings are measured in a clockwise direction from the North.

11. (a) Interquartile range $= 235 - 150$

$= \mathbf{85\ g}$

EXAM TIP

Interquartile range = Upper quartile – lower quartile

(b) P(person's estimate is less than k grams) $= \frac{3}{5}$

Number of people whose estimate is less than k grams

$= \frac{3}{5} \times 80$

$= 48$

From the graph, $k = \mathbf{220}$.

EXAM TIP

A common error is in reading off the frequency axis at $\frac{2}{5} \times 80 = 32$ to obtain the value of k.

12. (a) When $a = 98.31$, $b = 18.31$ and $c = 0.361$,

$$D = \frac{98.31}{18.31} - 18.31(0.361)^2$$

$= \mathbf{3.0}$ (to 2 s.f.)

(b)

$$D = \frac{a}{b} - bc^2$$

$$bD = a - b^2c^2$$

$$b^2c^2 = a - bD$$

$$c^2 = \frac{a - bD}{b^2}$$

$$\boldsymbol{c = \pm\sqrt{\frac{a - bD}{b^2}}}$$

EXAM TIP

Make c^2 the subject of the formula, then take the square root on both sides of the equation.

13. (a) $\angle OBA = \angle OAB$ (base $\angle$s of isos. $\triangle$)

$= 28°$

$\angle ABC = 90°$ (rt. $\angle$ in semicircle)

$\therefore \angle OBC = 90° - 28°$

$= \mathbf{62°}$

EXAM TIP

Alternatively, apply angle at centre = 2 × angle at circumference to find $\angle BOC$, before applying base angles of isosceles $\triangle OBC$.

(b) $\angle ACD = 180° - 60° - 28°$ ($\angle$ sum of $\triangle$)

$= 92°$

Since $\angle ACD \neq 90°$, then by the converse of tangent perpendicular to radius, DC is not a tangent to the circle.

EXAM TIP

DC is a tangent to the circle at C if $\angle OCD = 90°$.

14. $6x + 5y = 2$ —— (1)

$10x - 4y = 65$ —— (2)

(1) × 4: $24x + 20y = 8$ —— (3)

(2) × 5: $50x - 20y = 325$ —— (4)

(3) + (4): $74x = 333$

$x = 4\frac{1}{2}$

Substitute $x = 4\frac{1}{2}$ into (1):

$$6\left(4\frac{1}{2}\right) + 5y = 2$$

$$27 + 5y = 2$$

$$5y = -25$$

$$y = -5$$

$\boldsymbol{\therefore x = 4\frac{1}{2}, y = -5}$

EXAM TIP

Use the method of elimination or substitution.

15. $\angle CBA : \angle BAD = 9 : 3$

$= 3 : 1$

$\angle BAD = \frac{1}{3+1} \times 180°$ (int. $\angle$s, BC // AD)

$= 45°$

$\angle ADC = \frac{2}{3} \times 45°$

$= 30°$

$\therefore \angle BCD = 180° - 30°$ (int. $\angle$s, BC // AD)

$= \mathbf{150°}$

EXAM TIP

$\angle CBA + \angle BAC = 180°$ and $\angle BCD + \angle ADC = 180°$.

16. Let \$$P$ be the sum of money invested.

$$P + 2132.16 = P\left(1 + \frac{3.2}{100}\right)^5$$

$= 1.1706P$ (to 5 s.f.)

$0.1706P = 2132.16$

$P = 12\,499.99$ (to the nearest cent)

$= 12\,500$ (to the nearest dollar)

$\therefore$ Alice invested **\$12 500**.

EXAM TIP

When using the formula for compound interest, $P\left(1 + \frac{3.2}{100}\right)^5$ gives the total amount after 5 years, not the interest earned in 5 years.

17. **(a)** $n((A' \cup B) \cap (A \cup B')) = 3 + 9$
$= \mathbf{12}$

EXAM TIP

$(A' \cup B)$ refers to the regions not in A or the regions in B. $(A \cup B')$ refers to the regions in A or the regions not in B. The intersection refers to the regions that are common.

(b) $n((A \cap B') \cup (A \cup B)') = 7 + 9$
$= \mathbf{16}$

EXAM TIP

$(A \cap B') \cup (A \cup B)'$ refers to the region in A but not in B or the region not in either A or B.

18. **(a)** $263 = \mathbf{2.63 \times 10^2}$

(b) **(i)** $3.4 \times 10^{99} = \mathbf{0.34 \times 10^{100}}$

(ii) $(4.7 \times 10^{100}) + (3.4 \times 10^{99})$
$= (4.7 \times 10^{100}) + (0.34 \times 10^{100})$
$= \mathbf{5.04 \times 10^{100}}$

EXAM TIP

A number is expressed in standard form when written as $A \times 10^n$, where n is an integer and $1 \leqslant A < 10$.

19. $\text{HCF} = 2^4 \times 5$
$\text{LCM} = 2^6 \times 3^2 \times 5^3$
$720 = 2^4 \times 3^2 \times 5$
$\therefore N = 2^6 \times 5^3$
$= \mathbf{8000}$

EXAM TIP

Compare the powers of 2, 3 and 5. 3^2 (or 3) is not a factor of N because it is not a factor of the HCF.

20. $\sin(2x°) = 0.561$

$2x = 34.125°$ (to 3 d.p.) or $2x = 180° - 34.125°$
$x = 17.1°$ (to 1 d.p.) $\quad = 145.875°$ (to 3 d.p.)
$x = 72.9°$ (to 1 d.p.)

$\therefore x = \mathbf{17.1°}$ or $\mathbf{72.9°}$

EXAM TIP

Recall: $\sin A = \sin(180° - A)$.

21. **(a)** $\text{Mean} = \dfrac{12 + 24 + 8 + 21 + 28 + 17 + 2p + 4p^2}{8}$

$= \dfrac{4p^2 + 2p + 110}{8}$

$= \dfrac{2p^2 + p + 55}{4}$

Given that mean = 17.5,

$\dfrac{2p^2 + p + 55}{4} = 17.5$

$2p^2 + p + 55 = 70$
$2p^2 + p - 15 = 0$
$(2p - 5)(p + 3) = 0$
$2p - 5 = 0$ or $p + 3 = 0$
$p = 2.5$ $\quad p = -3$

Since $p > 0$, then $p = 2.5$. (shown)

(b)

	Mean	Standard deviation
Li	17.5	7.89
Li's sister	20	9.51

1. Li generally received fewer text messages than his sister as his mean (17.5) is less than that of his sister (20).
2. The number of text messages Li received has a smaller spread (lower standard deviation of 7.89 against 9.51) and is more clustered around the mean.

22. $(2x - 3)(3x^2 + 2x - 5) = 6x^3 + 4x^2 - 10x - 9x^2 - 6x + 15$
$= \mathbf{6x^3 - 5x^2 - 16x + 15}$

23. **(a)** Using Pythagoras' Theorem,

$y^2 + (3x - 1)^2 = (3x + 4)^2$
$y^2 = (3x + 4)^2 - (3x - 1)^2$
$= 9x^2 + 24x + 16 - 9x^2 + 6x - 1$
$= 30x + 15$
$= 15(2x + 1)$

Since $2x$ is a multiple of 2, then $2x + 1$ is an odd number, and $15(2x + 1)$ must be an odd number.

The square root of an odd number cannot be an even number.

$\therefore$ y is an odd number. (shown)

EXAM TIP

The square of an even number is an even number. The square of an odd number is an odd number.

(b) Let $y^2 = 15^2$.

Then $y = 15$.
$2x + 1 = 15$
$2x = 14$
$x = 7$

$\therefore$ A possible set of values is $\mathbf{x = 7, y = 15}$.

24. $\dfrac{3x^2 + 6x}{x^4 - 16} = \dfrac{3x(x + 2)}{(x^2 + 4)(x^2 - 4)}$

$= \dfrac{3x(x + 2)}{(x^2 + 4)(x + 2)(x - 2)}$

$= \mathbf{\dfrac{3x}{(x^2 + 4)(x - 2)}}$

EXAM TIP

$x^4 - 16 \neq (x^2 - 4)^2$

25. $x^2 - 12x + 17 = 0$
$(x - 6)^2 - 36 + 17 = 0$
$(x - 6)^2 = 19$
$x - 6 = \pm\sqrt{19}$
$x = 6 \pm \sqrt{19}$
$= \mathbf{10.36}$ or $\mathbf{1.64}$ (to 2 d.p.)

EXAM TIP

To complete the square, $x^2 + bx + c = \left(x + \dfrac{b}{2}\right)^2 - \left(\dfrac{b}{2}\right)^2 + c$.

26. **(a)** Volume of cone = Volume of cylinder

$\dfrac{1}{3}\pi(7.5)^2 h = \pi(7.5)^2(20)$

$h = \mathbf{60}$

EXAM TIP

Volume of cone $= \frac{1}{3}\pi r^2 h$; volume of cylinder $= \pi r^2 h$

(b) Volume of cylinder $= \pi(10)^2(25)$

$= 2500\pi \text{ cm}^3$

Volume of cone $= \frac{1}{3}\pi(10)^2(90)$

$= 3000\pi \text{ cm}^3$

Volume of water $= \frac{1}{2}(2500\pi + 3000\pi)$

$= 2750\pi \text{ cm}^3$

$$\frac{\text{Volume of empty space}}{\text{Volume of cone}} = \frac{2750\pi}{3000\pi}$$

$$\left(\frac{d}{90}\right)^3 = \frac{2750\pi}{3000\pi}$$

$$= \frac{11}{12}$$

$$\frac{d}{90} = \sqrt[3]{\frac{11}{12}}$$

$$d = 90 \times \sqrt[3]{\frac{11}{12}}$$

$= 87.4$ (to 3 s.f.)

$\therefore$ Depth of empty space $=$ **87.4 cm**

EXAM TIP

Since the cones are similar, apply $\frac{V_1}{V_2} = \left(\frac{h_1}{h_2}\right)^3$.

27. (a) $\overrightarrow{BC} = \frac{2}{3}\overrightarrow{BA}$

$= \frac{2}{3}\left(\overrightarrow{OA} - \overrightarrow{OB}\right)$

$= \frac{2}{3}(3\mathbf{a} - 6\mathbf{b})$

$= 2\mathbf{a} - 4\mathbf{b}$

$\therefore \overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC}$

$= 6\mathbf{b} + 2\mathbf{a} - 4\mathbf{b}$

$= 2\mathbf{a} + 2\mathbf{b}$ (shown)

(b) $\overrightarrow{OP} = \frac{1}{2}\overrightarrow{OC}$

$= \frac{1}{2}(2\mathbf{a} + 2\mathbf{b})$

$= \mathbf{a} + \mathbf{b}$

$\overrightarrow{OQ} = \overrightarrow{OA} + \overrightarrow{AQ}$

$= \overrightarrow{OA} + m\overrightarrow{AP}$

$= \overrightarrow{OA} + m\left(\overrightarrow{OP} - \overrightarrow{OA}\right)$

$= 3\mathbf{a} + m(\mathbf{a} + \mathbf{b} - 3\mathbf{a})$

$= 3\mathbf{a} - 2m\mathbf{a} + m\mathbf{b}$

$= (3 - 2m)\mathbf{a} + m\mathbf{b}$ ——— (1)

$\overrightarrow{OQ} = n\overrightarrow{OB}$

$= 6n\mathbf{b}$ ——— (2)

Equating coefficients of **a**,

$3 - 2m = 0$

$m = 1.5$

Equating coefficients of **b**,

$6n = 1.5$

$n = 0.25$

$\therefore OQ : QB = n : 1 - n$

$= 0.25 : 0.75$

$= \mathbf{1 : 3}$

EXAM TIP

Form two expressions for $\overrightarrow{OQ}$, and solve for m and n first.

(c) $\frac{\text{Area of } \triangle OAC}{\text{Area of } \triangle OBC} = \frac{1}{2}$

$\frac{\text{Area of } \triangle OAC}{25} = \frac{1}{2}$

Area of $\triangle OAC = \frac{1}{2} \times 25$

$= \mathbf{12.5 \text{ cm}^2}$

EXAM TIP

$\triangle OAC$ and $\triangle OBC$ are not similar triangles. They have the same height, so the areas are proportional to the lengths of the bases.

Paper 2 — October/November 2024

1. (a) $-4 \leqslant 3 + 2x < 5$

$-4 \leqslant 3 + 2x$ and $3 + 2x < 5$

$-2x \leqslant 7$ $\quad$ $2x < 2$

$x \geqslant -3.5$ $\quad$ $x < 1$

$\therefore -3.5 \leqslant x < 1$

$\therefore$ The integers are **−3, −2, −1** and **0**.

EXAM TIP

Solve $-4 \leqslant 3 + 2x$ and $3 + 2x < 5$.

(b) $\frac{20p^3q}{7r^2} \div \frac{8p^2r}{21q^4} = \frac{20p^3q}{7r^2} \times \frac{21q^4}{8p^2r}$

$$= \mathbf{\frac{15pq^5}{2r^3}}$$

(c) $\left(\frac{a^9}{64b^{12}}\right)^{-\frac{2}{3}} = \left(\frac{64b^{12}}{a^9}\right)^{\frac{2}{3}}$

$$= \left(\frac{4^3b^{12}}{a^9}\right)^{\frac{2}{3}}$$

$$= \mathbf{\frac{16b^8}{a^6}}$$

EXAM TIP

Recall: $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$.

(d) $\frac{x}{x+2} + \frac{3}{4-x} - 1 = \frac{x(4-x) + 3(x+2) - (x+2)(4-x)}{(x+2)(4-x)}$

$$= \frac{4x - x^2 + 3x + 6 - (4x - x^2 + 8 - 2x)}{(x+2)(4-x)}$$

$$= \frac{4x - x^2 + 3x + 6 - 4x + x^2 - 8 + 2x}{(x+2)(4-x)}$$

$$= \mathbf{\frac{5x - 2}{(x+2)(4-x)}}$$

EXAM TIP

Combine the fractions into a single fraction with denominator $(x + 2)(4 - x)$.

2. (a) (i) $\mathbf{T} = \begin{pmatrix} \mathbf{12} \\ \mathbf{15} \end{pmatrix}$

(ii) $\mathbf{M} = \mathbf{CT}$

$$= \begin{pmatrix} 120 & 40 \\ 90 & 30 \\ 50 & 20 \end{pmatrix} \begin{pmatrix} 12 \\ 15 \end{pmatrix}$$

$$= \begin{pmatrix} 120 \times 12 + 40 \times 15 \\ 90 \times 12 + 30 \times 15 \\ 50 \times 12 + 20 \times 15 \end{pmatrix}$$

$$= \begin{pmatrix} \mathbf{2040} \\ \mathbf{1530} \\ \mathbf{900} \end{pmatrix}$$

EXAM TIP

The product of a (3×2) matrix and a (2×1) matrix yields a (3×1) matrix.

(iii) Difference $= \left(\frac{9}{10} - \frac{3}{5}\right) \times \$(2040 + 1530 + 900)$

$= \mathbf{\$1341}$

(b) Number of tickets sold in the second week of the month

$= (100\% - 56\%) \times 5075$

$= \mathbf{2233}$

(c) afternoon tickets : evening tickets : night tickets

3	:	8		
		6	:	11
9	:	24	:	44

$\therefore$ afternoon tickets : night tickets = **9 : 44**

EXAM TIP

Find the LCM of 8 and 6.

3. (a) L: $4y = 3x - 2$ ——— (1)

M: $2y = kx + 13$ ——— (2)

Substitute $x = -4$, $y = p$ into (1):

$4p = 3(-4) - 2$

$= -14$

$p = -3.5$

Substitute $x = -4$, $y = -3.5$ into (2):

$2(-3.5) = k(-4) + 13$

$-7 = -4k + 13$

$4k = 20$

$k = 5$

$\therefore$ $\boldsymbol{k = 5, p = -3.5}$

EXAM TIP

$x = -4$ and $y = p$ satisfy both equations of L and M.

(b) Gradient of $XY = -\frac{2}{5}$

Substitute $x = -3$, $y = 7$, $m = -\frac{2}{5}$ into $y = mx + c$:

$7 = -\frac{2}{5}(-3) + c$

$= \frac{6}{5} + c$

$c = \frac{29}{5}$

$\therefore$ Equation of XY: $y = -\frac{2}{5}x + \frac{29}{5}$, i.e. $\mathbf{5y = 29 - 2x}$

EXAM TIP

If $\overrightarrow{XY} = \begin{pmatrix} h \\ k \end{pmatrix}$, then the gradient of a line passing through X and Y is $\frac{k}{h}$.

(c) Given that $AB = 25$ units,

$\sqrt{(15 - 8)^2 + (11 - a)^2} = 25$

$(15 - 8)^2 + (11 - a)^2 = 625$

$49 + 121 - 22a + a^2 = 625$

$a^2 - 22a - 455 = 0$

$(a - 35)(a + 13) = 0$

$a = 35$ or $a = -13$

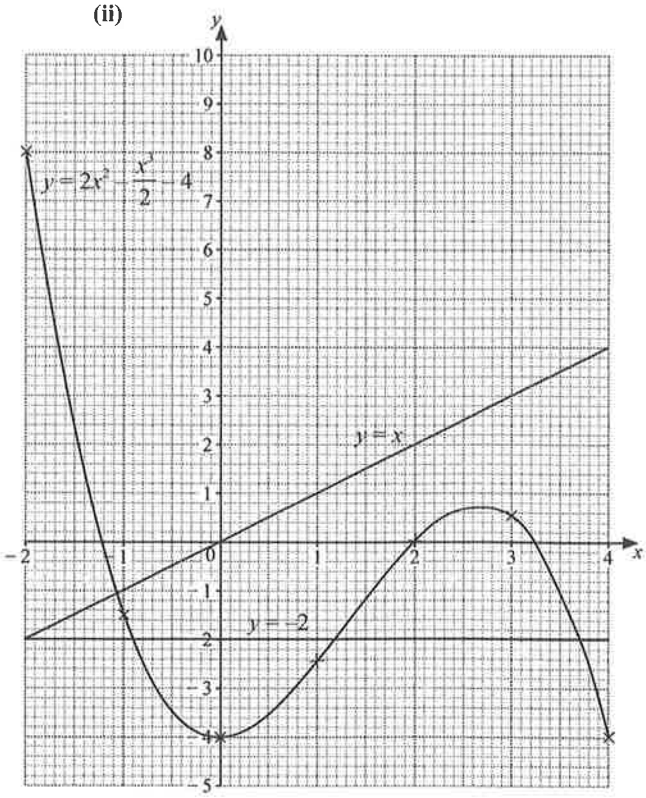
Since $a < 0$, then $a = \mathbf{-13}$.

4. (a) (i) When $x = -2$,

$y = 2(-2)^2 - \frac{(-2)^3}{2} - 4$

$= \mathbf{8}$

(ii)



(iii) The line $y = x$ intersects the curve $y = 2x^2 - \frac{x^3}{2} - 4$ at exactly one point, so the equation $2x^2 - \frac{x^3}{2} - 4 = x$ has only one solution.

(iv) $4x^2 - x^3 - 4 = 0$

$2x^2 - \frac{x^3}{2} - 2 = 0$

$2x^2 - \frac{x^3}{2} - 4 = -2$

Draw $y = -2$.

From the graph, $x = \mathbf{-0.9}$, $x = \mathbf{1.2}$ or $x = \mathbf{3.7}$.

EXAM TIP

The solutions correspond to the x-coordinates of the points of intersection between $y = 2x^2 - \frac{x^3}{2} - 4$ and $y = -2$.

(b) $y = ka^x$

When $x = 0$, $y = 5$,

$5 = k(a^0)$

$k = 5$

Substitute $x = 3$, $y = 135$ into $y = 5a^x$:

$135 = 5a^3$

$a^3 = 27$

$a = 3$

Substitute $x = 4$, $y = p$ into $y = 5(3^x)$:

$p = 5(3^4)$

$= 405$

$\therefore p = \mathbf{405}$

EXAM TIP

Recall: $a^0 = 1$.

5. **(a)** Using Cosine Rule,

$405^2 = 175^2 + 280^2 - 2(175)(280) \cos \angle BAC$

$164\ 025 = 109\ 025 - 98\ 000 \cos \angle BAC$

$98\ 000 \cos \angle BAC = -55\ 000$

$\cos \angle BAC = -\frac{55}{98}$

$\angle BAC = 124.1°$ (to 1 d.p.) (shown)

EXAM TIP

Apply the Cosine Rule: $BC^2 = AB^2 + AC^2 - 2(AB)(AC) \cos \angle BAC$.

(b) $\angle DAC = 360° - 90° - 48° - 124.141°$ ($\angle$s at a pt.)

$= 97.859°$ (to 3 d.p.)

$\angle ADC = 180° - 32° - 97.859°$ ($\angle$ sum of $\triangle$)

$= 50.141°$ (to 3 d.p.)

Using Sine Rule,

$\frac{AD}{\sin 32°} = \frac{280}{\sin 50.141°}$

$AD = \frac{280 \sin 32°}{\sin 50.141°}$

$= \mathbf{193\ m}$ (to 3 s.f.)

EXAM TIP

Apply the Sine Rule: $\frac{AD}{\sin \angle ACD} = \frac{AC}{\sin \angle ADC}$.

(c) $\tan 34° = \frac{AT}{280}$

$AT = 280 \tan 34°$

$= 188.86$ m (to 5 s.f.)

$\tan \angle TBA = \frac{188.86}{175}$

$\angle TBA = 47.2°$ (to 1 d.p.)

$\therefore$ Angle of elevation of T from $B = \mathbf{47.2°}$

EXAM TIP

Consider a right-angled triangle and use the tangent ratio to find the angle of elevation.

6. **(a)** Time taken for the first 40 km $= \frac{40}{x}$ h

Time taken for the remaining 20 km $= \frac{20}{x - 30}$ h

Given that the total time taken is 50 min $= \frac{5}{6}$ h,

$\frac{40}{x} + \frac{20}{x - 30} = \frac{5}{6}$

$240(x - 30) + 120x = 5x(x - 30)$

$240x - 7200 + 120x = 5x^2 - 150x$

$5x^2 - 510x + 7200 = 0$

$x^2 - 102x + 1440 = 0$ (shown)

EXAM TIP

Since Nadia's average speed on the local roads is 30 km/h slower, then her speed on the local roads is $(x - 30)$ km/h, not $(x + 30)$ km/h.

(b) $x^2 - 102x + 1440 = 0$

$x = \frac{-(-102) \pm \sqrt{(-102)^2 - 4(1)(1440)}}{2(1)}$

$= \frac{102 \pm \sqrt{4644}}{2}$

$= \mathbf{85.07}$ or $\mathbf{16.93}$ (to 2 d.p.)

(c) The solution $x = 16.93$ must be rejected, as it would result in the time taken by Nadia to drive on local roads, $\frac{20}{x - 30}$ h, to be a negative value.

(d) Difference in times $= \left(\frac{40}{85.073} - \frac{20}{85.073 - 30}\right)$ h

$= 0.107\ 03$ h

$= 6.4218$ min

$= \mathbf{6\ min\ 25\ s}$ (to the nearest second)

7. **(a)** **(i)** $\frac{120}{20} = 60$

Median positions: 60th and 61st

$\therefore$ Interval that contains the median mass:

$\mathbf{120 < m \leqslant 140}$

EXAM TIP

Median = Mean of the two middle values when the number of data values is even

(ii) Estimated mean mass

$$= \frac{8(70) + 21(90) + 27(110) + 46(130) + 18(150)}{120}$$

$= \mathbf{117.5\ g}$

EXAM TIP

To find an estimate for the mean mass, find the mid-value of each interval first.

(iii) Estimated standard deviation

$$= \sqrt{\frac{8(70)^2 + 21(90)^2 + 27(110)^2 + 46(130)^2 + 18(150)^2}{120} - 117.5^2}$$

$= \mathbf{22.7\ g}$ (to 3 s.f.)

(iv) The mean mass of the 80 apples added to the box is less than 110 g.

(b) **(i)** The maximum and minimum masses might not be 180 g and 80 g respectively, as the histogram groups the masses into intervals of 20 g.

(ii) P(mass between 120 g and 160 g) $= \frac{32 + 48}{150}$

$= \mathbf{\frac{8}{15}}$

(iii) P(two have a mass less than 120 g, one has a mass greater than 160 g)

$= \frac{30}{150} \times \frac{29}{149} \times \frac{40}{148} \times 3$

$= \frac{174}{5513}$

$= \mathbf{0.0316}$ (to 3 s.f.)

8. **(a)** **(i)** $OM = ON$ (equal chords)

$OB = OC$ (radii)

$\angle OMB = 90°$ ($\perp$ bisector of chord)

$\angle ONC = 90°$ ($\perp$ bisector of chord)

$\therefore \angle OMB = \angle ONC$

$\therefore \triangle BMO$ is congruent to $\triangle CNO$. (Right angle-hypotenuse-side Congruence Test)

EXAM TIP

Apply the geometrical properties of circles to show that the two triangles are congruent.

One other pair of equivalence is $BM = CN$, which is derived from equal chords $AM = CD$, and M and N being the respective midpoints.

(ii) $\angle AOB = \angle COD = 1.8$ rad

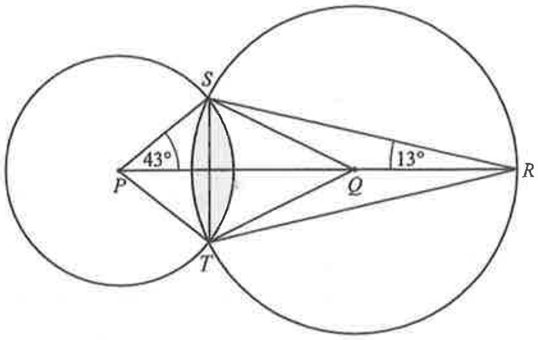
Length of major arc $AB = 4(2\pi - 1.8)$

$= \mathbf{17.9\ cm}$ (to 3 s.f.)

EXAM TIP

Arc length $= r\theta$, where θ is in radians

(b)



Area of shaded segment formed between ST and arc of circle centre P

$= \frac{86°}{360°} \times \pi(4.5)^2 - \frac{1}{2}(4.5)(4.5) \sin 86°$

$= 5.0971\ \text{cm}^2$ (to 5 s.f.)

$\angle SQP = 2(13°)$ ($\angle$ at centre $= 2\ \angle$ at circumference)

$= 26°$

Area of shaded segment formed between ST and arc of circle centre Q

$= \frac{52°}{360°} \times \pi(7)^2 - \frac{1}{2}(7)(7) \sin 52°$

$= 2.9292\ \text{cm}^2$ (to 5 s.f.)

$\therefore$ Shaded area $= 5.0971 + 2.9292$

$= \mathbf{8.03\ cm^2}$ (to 3 s.f.)

EXAM TIP

Area of segment = Area of sector – area of triangle

9. **(a)** Total mass $= 0.765 \times 5^3$

$= \mathbf{95.625\ g}$

(b) Let the mass of wax used be m g.

$m + \frac{8}{100} \times m = 150$

$1.08m = 150$

$m = 139$ (to 3 s.f.)

$\therefore$ Mass of wax used is 139 g (shown)

(c)

	Smaller cylinder	Larger cylinder
Radius	2 cm	3 cm
Height	8 cm	12 cm
Volume	$\pi(2)^2(8) = 32\pi\ \text{cm}^3$	$\pi(3)^2(12) = 108\pi\ \text{cm}^3$
Total mass of candle	$0.765 \times 32\pi = 24.48\pi$ g	$0.765 \times 108\pi = 82.62\pi$ g
Mass of wax	$\frac{100}{108} \times 24.48\pi$ $= 71.209$ g (to 5 s.f.)	$\frac{100}{108} \times 82.62\pi$ $= 240.33$ g (to 5 s.f.)
Mass of fragrance oil	$\frac{8}{100} \times 71.209$ $= 5.6968$ g (to 5 s.f.)	$\frac{8}{100} \times 240.33$ $= 19.227$ g (to 5 s.f.)
Volume of fragrance oil	5.6968 ml (to 5 s.f.)	19.227 ml (to 5 s.f.)
Length of wick	8.5 cm	12.5 cm

Assume Wei makes 10 candles of each size to sell.

Total mass of wax needed $= 10(71.209 + 240.33)$

$= 3115.4$ g (to 5 s.f.)

Total volume of fragrance oil needed
$= 10(5.6968 + 19.227)$
$= 249.24$ ml (to 5 s.f.)

Cost of supplies:
Wax: $3(\$26) + \$16 = \$94$
Fragrance oil: $4(\$22) + \$12 = \$100$
Candle wick: $2(\$8) = \$16$
Total cost $= \$94 + \$100 + \$16$
$= \$210$

$$\frac{\text{Volume of smaller candle}}{\text{Volume of larger candle}} = \left(\frac{4}{6}\right)^3 = \frac{8}{27}$$

Estimated cost of smaller candle $= \frac{8}{35} \times \$210 \times \frac{1}{10}$
$= \$4.80$

Estimated cost of larger candle $= \frac{27}{35} \times \$210 \times \frac{1}{10}$
$= \$16.20$

Assume Wei wants to make a profit of 50%.

Suggested price of smaller candle $= \frac{150}{100} \times \$4.80$
$= \mathbf{\$7.20}$

Suggested price of larger candle $= \frac{150}{100} \times \$16.20$
$= \mathbf{\$24.30}$

Paper 1 October/November 2023

1. Amount Sima receives $= \frac{9}{9+7} \times \$480$
$= \mathbf{\$270}$

Amount Ken receives $= \$480 - \$270$
$= \mathbf{\$210}$

2. **(a)** $\sin x° = 0.9301$
$x° = \sin^{-1} 0.9301$ or $x° = 180° - \sin^{-1} 0.9301$
$= 68.5°$ (to 1 d.p.) $\quad = 111.5°$ (to 1 d.p.)
$\therefore x = \mathbf{68.5}$ or $\mathbf{111.5}$

EXAM TIP
Recall: $\sin A = \sin (180° - A)$.

(b) π rad $= 180°$
1.36 rad $= \frac{180°}{\pi} \times 1.36$
$= \mathbf{77.9°}$ (to 1 d.p.)

3. **(a)** $(2c^3d^2)^3 = \mathbf{8c^9d^6}$

EXAM TIP
Recall: $(a^m)^n = a^{mn}$.

(b) $16 \times 5^4 + 9 \times 25^2 = 16 \times 5^4 + 9 \times (5^2)^2$
$= 16 \times 5^4 + 9 \times 5^4$
$= 5^4 \times (16 + 9)$
$= 5^4 \times 25$
$= 5^4 \times 5^2$
$= 5^{4+2}$
$= 5^6$
$\therefore k = \mathbf{6}$

4. **(a)** $M = p^{16} \times q^8 \times r^{12}$
Since the indices of the prime factors of M are positive even numbers, then $M = p^{16} \times q^8 \times r^{12} = (p^8 \times q^4 \times r^6)^2$, i.e. M is a perfect square.

EXAM TIP
The powers of the prime factors of a perfect square are multiples of 2.

(b) **(i)** For $N = S \times k$ to be a perfect cube, smallest possible integer value of $k = \mathbf{2 \times 7^2}$.

EXAM TIP
For $N = S \times k$ to be a perfect cube, the powers of its prime factors must be multiples of 3.

(ii) LCM of S and $T = \mathbf{2^{20} \times 7^{10} \times 11^{15}}$

5. **(a)** The first year's interest is more than 3% of \$4500 because the interest is compounded every month, so the principal increases every month.

EXAM TIP
The first year's interest is 3% of \$4500 only if simple interest rate is used.
Since the interest is compounded monthly, then the first year's interest is $\left[\left(1 + \frac{0.25}{100}\right)^{12} - 1\right]$ of \$4500, which is approximately 3.04% of \$4500.

(b) Value of investment $= \$4500\left(1 + \frac{0.25}{100}\right)^{12}$
$= \mathbf{\$4636.87}$ (to 2 d.p.)

6. **(a)** $x^2 - 14x + b = (x - 7)^2 - 7^2 + b$
$= (x - 7)^2 + b - 49$
By comparing,
$a = -7$
$b - 49 = -25$
$b = 24$
$\therefore \mathbf{a = 7, b = 24}$

EXAM TIP
Complete the square.

(b) Equation of line of symmetry: $x = \mathbf{7}$

7. **(a)** Number of stamps now $= 118\% \times 550$
$= \frac{118}{100} \times 550$
$= \mathbf{649}$